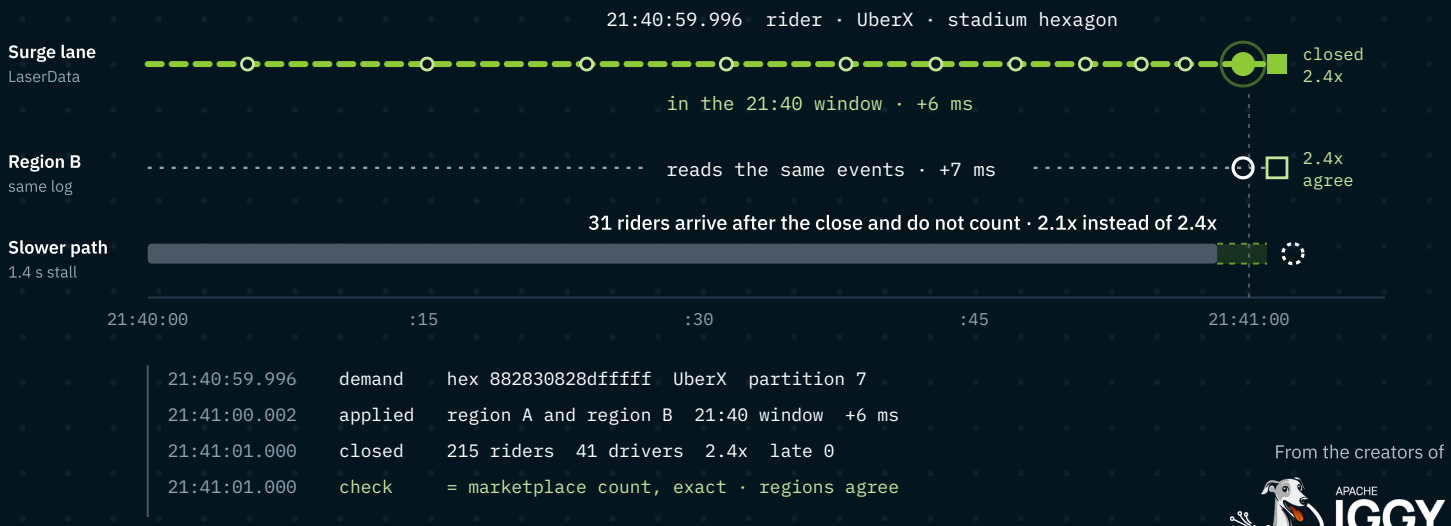
laserdata

Prepared for **Uber**

Real-time surge windows for Uber, on LaserData.

Every rider and driver event reaches **the surge window that owns its hexagon in milliseconds, so late events stop dropping out of the multiplier**. It runs beside Kafka and Flink, so nothing else has to move.

One hexagon in one window,
as the event log sees it.
Now multiply by 5M hexagons.




```
21:40:59.996   demand    hex 882830828dfffff  UberX  partition 7
21:41:00.002   applied   region A and region B  21:40 window  +6 ms
21:41:01.000   closed    215 riders  41 drivers  2.4x  late 0
21:41:01.000   check     = marketplace count, exact · regions agree
```

From the creators of APACHE IGGY

15×
faster P99 latency
than Apache Kafka

4×
higher throughput
than Apache Kafka

60–70%
lower infrastructure cost
compute, storage and network

Apache Iggy vs Apache Kafka, LaserData testing. Results vary by workload.

What you shared	Surge computes multipliers per hexagon in a time window, from Kafka through Flink, in every region. **"The late-arriving messages do not contribute to the surge computation."** 120k events/s into per-hexagon features, about 5M hexagons. (Uber, SIGMOD 2021; Uber Engineering, Aug 2021)
Where it fits	**The surge window lane:** demand and supply events keyed by hexagon into each region's window, every window checked exact. Kafka, Flink and the pricing store stay.
Next	**The demand and supply feature windows** on the same lane, rebuilt from the log on restart.
Underneath	**One durable event log.** LaserData, powered by Apache Iggy.

Try it in 15 minutes. Let the numbers decide.

A ready-made kit for Uber: **4 cities, 4,000 hexagons, 45 demand events per hexagon a minute**, 6,000 drivers, 60 s windows in two regions, a stadium spike, an analytics flood and a Kafka shadow bridge. It prints your lane's p99 and windows checked exact every second.

Open the Uber kit and write-up →

Then a 2–4 week evaluation on one city's surge inputs, beside Kafka. **No advantage, no migration.**

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